Software Engineering Project Jan-2026

Milestone 2 Deliverables for IIT BS Drama Society Platform

Submitted By

- Team Name: ServiceSync
- Team Code: Team-070
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Project Details

- Software Name: Aayam (Cultural Club Operations Platform)
- Department Context: IIT BS Cultural & Drama Club Department
- System Type: Community Engagement, Resource Allocation & Club Management Platform
- Date of Submission: 28-06-2026

Team Details

Name	Roll Number	Role
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Design of Components

1. Authentication & Access Management Component

- Handle user registration with email, password, and member ID validation
- Authenticate users using JWT access tokens with secure login and logout (Redis blacklisting fallback)
- Fetch and update user profiles, including personal details, interests, and profile pictures
- Differentiate privileges using a role-based access control (Admin, Member, etc.)

2. Label & User Grouping Component

- Create and manage dynamic labels/tags (e.g., skillset, interest group, VIP) and metadata
- Assign and remove labels from specific users for targeted filtering
- Fetch list of users based on their assigned custom labels (for dashboards)

3. Events, Plays & Auditions Component

- Fetch available plays, ongoing events, and open casting/crew positions
- Submit audition requests or participation registrations for specific open roles
- Create and update backend events and manage associated schedules

4. Stage & Venue Booking Component

- View real-time availability of the stage/venue across various time slots (morning, afternoon, evening)
- Submit primary or secondary stage booking requests with details
- Ensure isolated handling of stage requests internally for admins

5. Operations & Core Dashboard Component

- Generate personalized dashboard views for users based on their permissions and activity
- Aggregate active bookings, upcoming events, and user notifications in one place
- Support fetch and update of personal daily/weekly schedules (`get_my_schedule`)

6. Revenue & Metrics Tracking Component

- Maintain and update real-time revenue metrics like playbills sold and expected income
- Provide up-to-date business KPIs and insights for dashboard visualizations
- Distinguish corporate patrons from individual metrics for analytical reporting

7. Media & Gallery Component

- Aggregate event-specific media files (images, videos, etc.) uploaded by users or admins
- Facilitate media item creation, approval status updates, and soft/hard deletions
- Attach gallery content to events for historical documentation and discovery

8. Proposals Component

- Receive and track user-submitted proposals (e.g. for new plays or initiatives)
- Approve, reject, or update the status and notes of submitted proposals
- Centralize pending requests for admin review before final confirmation

9. Real-time Notifications & Outreach Component

- Send and aggregate role-specific user notifications and alerts
- Track notification read/unread states and allow batch marking them as read
- Support automated fetching and status updates of user notifications without reloading pages
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10. AI Intelligence Component

10.1 Event Summarizer

- **Overview:** Parses raw unstructured text (e.g. event details, notes, announcements) to automatically extract key metadata and autofill the event request form.
- **Input:** Raw unstructured event text, notes, or plain text details.
- **Output:** Autofilled event request form fields (Title, Date, Time, Venue, Description, Category, Summary Highlights, Tags).
- **Target Users:** Event Organizers, Club Admins, Production Maintainers.
- **Key Benefits:** Eliminating manual data entry when creating events; speeds up event request submissions.

10.2 Key Point Filter (Event & Play Requests & Portfolio)

- **Overview:** Scans and extracts essential key points from event/play requests submitted by sponsors or organizers (allowing admins to quickly grasp what they want and how they want it) as well as member portfolios and requests.
- **Input:** Event/play request descriptions, sponsor proposal notes, member portfolio documents, raw request text.
- **Output:** Extracted key points summary, structured requirement highlights (what they want, deliverables, timeline/budget needs), matched skill and experience tags.
- **Target Users:** Club Admins, Production Managers, Review Committees, Sponsors, Casting Directors.
- **Key Benefits:** Accelerates request and proposal review; provides instant clarity on sponsor/event expectations and candidate qualifications without reading lengthy descriptions.

10.3 Cue Partner & Line Reader

- **Overview:** Interactive rehearsal assistant that fetches scripts directly from the platform's Script Vault, speaks prompt lines for co-characters, and acts as a virtual cue partner for actors practicing solo.
- **Input:** Selected script from site Script Vault, target character role, actor's spoken response (audio microphone input).
- **Output:** Synthesized audio line delivery (cue lines), active line highlight in script UI, line accuracy feedback.
- **Target Users:** Actors, Cast Members, Rehearsal Participants.
- **Key Benefits:** Allows solo rehearsals anywhere using platform scripts; improves line memorization and prompt timing.

10.4 Character Suggester for Plays (Discussion)

- **Overview:** Uses the complete play script as context to extract and generate key personality traits and characteristics for requested roles (e.g. hero's rival, mentor), while suggesting matching actors and facilitating director casting discussions.

- **Input:** Complete play script text context, requested character query/prompt (e.g. "Rival of the hero"), member directory profiles.
- **Output:** Generated character trait breakdown (personality traits, relationship dynamics, narrative role), ranked actor recommendations, collaborative casting discussion thread.
- **Target Users:** Play Directors, Casting Committee, Production Admins.
- **Key Benefits:** Automated script analysis for character profiling; data-driven casting decisions and centralized director feedback.

10.5 Audition Sanity Checker

- **Overview:** Automated verification system that performs two core checks for audition submissions: (1) portfolio matching and scoring against applied role requirements, and (2) technical quality auditing of the provided audition video/audio.
- **Input:** Uploaded audition video/audio file, applicant member portfolio, target role criteria.
- **Output:** Dual audit report:
    - 1. **Portfolio Match Score:** Role compatibility score based on portfolio skills, past experience, and character criteria.
    - 2. **Audio/Video Quality Check:** Technical audit report (resolution, audio clarity/volume, lighting score, format/duration pass-fail status).
- **Target Users:** Auditioning Members, Performers, Casting Directors, Admins.
- **Key Benefits:** Guarantees applicant compatibility for applied roles; prevents poor-quality audio/video submissions

System Design – Class Diagram

Overview

The class diagram represents the complete system architecture of the **Aayam Drama Club Management System**, consisting of interconnected classes organized across multiple functional modules. It illustrates the relationships between core entities such as **User**, **Admin**, **Event**, **Play**, **Audition**, **Booking**, **Participation**, **Membership**, and **Volunteer**, providing a modular and scalable architecture for managing club operations, theatrical productions, and member engagement.

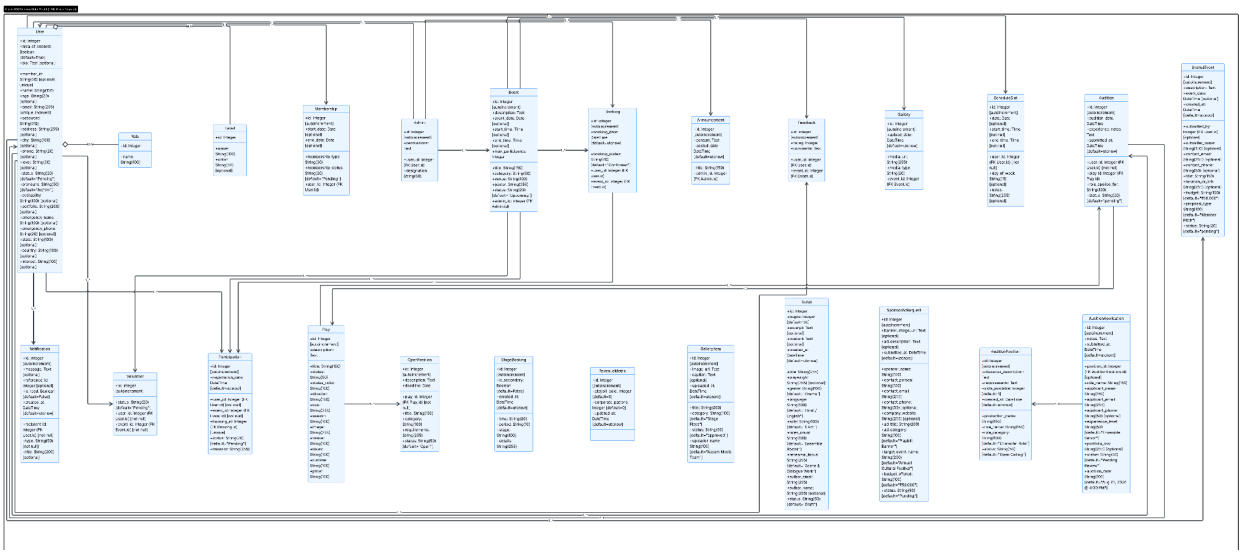
The design models the complete workflow of the drama club—from member registration and event management to auditions, production planning, volunteering, notifications, and feedback—while maintaining strong relationships between entities through a normalized database structure.

Key Highlights

- Covers **10 major modules** including User Management, Membership, Administration, Event Management, Production Management, Auditions, Scheduling, Media, Community Engagement, and Notifications.
- Clearly defines relationships between core entities such as **User**, **Admin**, **Event**, **Booking**, **Participation**, **Play**, **Audition**, **Membership**, **Volunteer**, and **Feedback**, accurately representing real-world drama club operations.
- Supports complete theatre workflows, including member registration, event organization, booking, participation tracking, volunteer management, auditions, role recruitment, production management, announcements, and media galleries.
- Implements **Role-Based Access Control (RBAC)** through a flexible User–Role relationship, enabling different permissions for administrators, members, volunteers, and other club roles.
- Follows a normalized database design, minimizing redundancy by separating business entities and connecting them through well-defined primary and foreign key relationships.
- Designed for scalability and modular development, allowing future extensions such as ticketing, costume management, rehearsal attendance, payment processing, sponsorship management, or AI-powered audition screening without major architectural changes.

Class Diagram

The following UML class diagram illustrates the structure of the system and the relationships between the primary classes.



Database Design – ER Diagram

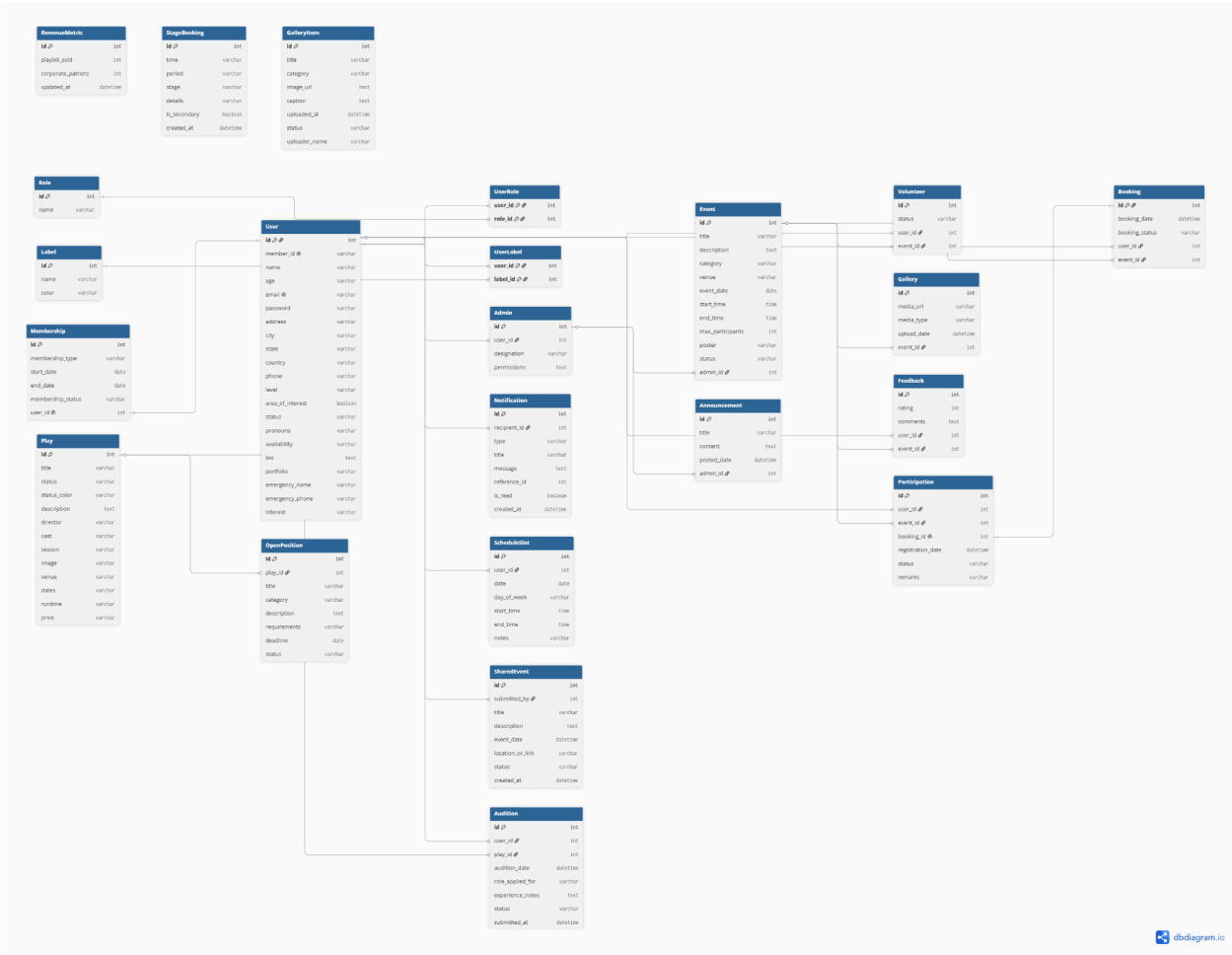
Overview

The ER (Entity-Relationship) diagram represents the database structure of the **Aayam (Cultural Club Operations Platform)**, defining entities, attributes, and relationships required for efficient data management. It captures how different components such as **User, UserRole, UserLabel, Label, Membership, Event**, and other entities are interconnected at the database level.

Key Highlights

- Defines relationships such as:
  - **One-to-Many:** User → Booking, Event → Booking, User → Feedback
  - **Many-to-One:** Booking → User, Booking → Event, Volunteer → Event
  - **Many-to-Many:** User ↔ Role, User ↔ Label
- Ensures data integrity and consistency across all modules.
- User-centric design, where all activities are linked back to a single user profile.
- Optimized for efficient querying and scalability.

ER Diagram



PROJECT SCHEDULE

Date	Phase / Focus	Main Agenda Points	Expected Outcome
Jun 23, 2026	Milestone 1 - Interview Prep & Insights	a. Formulate drama society interview questions; b. Set recording protocols; c. Interview Siddharth Sahu (Drama Secretary); d. Define initial project focus.	Interview completed and recorded, key pain points and workflow defined.
Jun 24, 2026	User Personas & Milestone Strategy	a. Define user tiers (Primary, Secondary, Regulatory); b. Establish research scope limits; c. Allocate analysis responsibilities; d. Plan Honor Code submission.	User groups categorized, tasks allocated for user analysis.
Jun 25, 2026	Document Standards & User Stories	a. Set Markdown format for master report; b. Define user stories for all tiers; c. Discuss misconduct and support routing; d. Draft manual transcript template.	User story guidelines established, initial transcript drafted.
Jun 27, 2026	Milestone 1 Closure & Git Strategy	a. Finalize Honor Code and video processing; b. Condense and format user stories; c. Define Git branching/deletion rules to avoid clutter.	Milestone 1 completed and verified for submission.
Jun 30, 2026	Milestone 2 Kickoff & UI Strategy	a. Submit collective team setup forms; b. Establish generic platform architecture; c. Design prompt workflows for UI uniformity; d. Set role-based financial visibility rules.	Milestone 2 kicked off, UI design rules and templates locked.
Jul 03, 2026	DB Schema & Architecture Alignment	a. Evaluate TypeScript/React framework; b. Plan database schema for scalability; c. Differentiate member vs. admin dashboard flows; d. Finalize design documentation structure.	Initial database schema mapped, UI branching protocol live.
Jul 07, 2026	Workflow Tracking & DB Refinement	a. Standardize central tracking on GitHub Kanban; b. Detail fields and relationships for database tables; c. Reconcile daily sync schedule to 9:00 PM.	Kanban tracking live, database fields defined professionally.
Jul 08, 2026	Code Reorganization	a. Isolate reusable sidebar and	UI codebase reorganized

Date	Phase / Focus	Main Agenda Points	Expected Outcome
	& Script Engine	role sub-folders; b. Integrate plays/scripts schema with SQLite; c. Map workflow for volunteering vs. event participation.	by roles, script tables integrated.
Jul 10, 2026	TA Review Prep & GenAI Scaffolding	a. Complete basic landing page and routes; b. Clean repository master branch; c. Outline 5 core GenAI features for future implementation.	Codebase cleaned, core dashboards ready for TA progress review.
Jul 15, 2026	GenAI Architecture & M2 Completion	a. Establish light LLM infrastructure; b. Design real-time roleplay audio loop (WebRTC/WebSockets); c. Map Swagger API documentation guidelines; d. Consolidate final design artifacts.	Milestone 2 design finalized, GenAI logic architected.
Aug 02, 2026	Milestone 3 (Sprint 1) Review	a. Deploy SQLite core schema tables and indices; b. Build out CRUD backend APIs for membership, events, and scripts; c. Implement free-tier LLM gateway.	Core APIs and database backend fully operational.
Aug 12, 2026	Milestone 4 (Sprint 2) Review	a. Integrate React frontend with developed REST backend; b. Configure global state routing for distinct roles; c. Conduct full integrated E2E system flow testing.	Complete integrated system application operational.
Aug 23, 2026	Milestone 5 (Final Submission)	a. Final verification and sanity check; b. Conduct dry-run presentation and record video demo; c. Compile master engineering report and finalize code zip. b. Conduct dry-run presentation and record video demo c. Compile master engineering report and finalize code zip	Project fully verified and submitted to drive.

Milestone 1 Plan (User Requirements & Research)

Duration: Jun 22 – Jun 28, 2026

Focus: Identify stakeholders, execute primary field research, extract core pain points, and define unified user stories.

- **Weekly Goals:**
    - Complete domain interviews with club stakeholders, synthesize user analysis tiers, establish repository infrastructure, and submit individual Honor Code files.
- **Major Activities:**
    - Conduct structural interview with Drama Society Secretary Siddharth Sahu to trace operational blockages.
    - Classify primary users (members, actors), secondary users (committee administrators), and tertiary/regulatory users (university admin, external sponsors).
    - Format data inputs into clean Markdown files (.md) to establish a single source of truth for downstream reports.
- **Deliverables:**
    - Processed interview audio recording trimmed and optimized to 1.25x playback speed.
    - Comprehensive Q&A text transcript alongside categorized user pain points.
    - Standardized user story log matching the "As a... I want to... so that..." formula.
    - Signed individual student Honor Code verification documents.
- **Expected Outcomes:**
    - Milestone 1 finalized, checked for alignment, and uploaded directly to the drive portal by the fixed Jun 28 deadline.

Milestone 2 Plan (System Design & Architecture)

Duration: Jun 29 – Jul 23, 2026

Focus: Define relational database structures, map role-specific visual components, and architect generative intelligence pipelines.

Week	Dates	Main Goals / Key Tasks	Focus Meeting Outcome
1	Jun 29 - Jul 05	- Kick off Milestone 2 planning parameters. - Define application scope around generic single-club architecture. - Separate component views for regular member and maintainer roles.	**Jul 03 Meeting:** Component framework chosen (TypeScript/React), SaaS scalability schema finalized, and layout folders mapped.
2	Jul 06 - Jul 12	- Map complete database relational fields. - Reorganize directory structure into dedicated subfolders. - Outline functional requirements for 5 core AI tools.	**Jul 07 & Jul 08 Meetings:** GitHub Kanban board setup live; schema upgraded to support plays/scripts Markdown files in SQLite.
3	Jul 13 - Jul 19	- Architect light LLM integration architecture. - Standardize free-tier API request management configurations. - Design low-latency interactive scripts loop blueprints.	**Jul 10 & Jul 15 Meetings:** Main branch code scrubbed, landing page complete, and GenAI threshold architecture locked.
4	Jul 20 - Jul 23	- Execute rigorous technical verification across all design artifacts. - Synthesize complete E-R diagram charts. - Assemble Milestone 2 PDF documentation.	**Jul 23 Target:** Complete structural blueprints, component screenshots, and architecture schemas validated for submission.

Sprint 1 (Milestone 3: Core Backend & API Development)

Duration: Jul 24 – Aug 02, 2026

Focus: Implement relational SQLite data models, build out core backend business endpoints, and establish generative gateways.

- **Backend Implementation:**
    - Construct core SQLite relational tables mapping the internal fields defined during design (Members, core roles, plays, and schedule timelines).
    - Incorporate index bindings for rapid lookup within active membership clusters and event logs.

- **API Development:**
    - Develop full CRUD REST routes managing membership states, application processes, and event logistics.
    - Document all operational routes explicitly matching Swagger/YAML visualization specifications discussed during technical sync.
- **Integration Tasks:**
    - Build out a proxy gateway managing requests toward Generative AI endpoints (Gemini/ChatGPT free tiers).
    - Embed request rate-limit filters to restrict processing strictly within the 15–30 requests per minute ceiling.
- **Testing:**
    - Write and run localized backend automated test suites to validate script handling stability and basic data input persistence.
- **Sprint Review & Deliverables:**
    - Backend service package containing clean database schemas and a documented Swagger file mapping operational API pathways.

Sprint 2 (Milestone 4: Frontend Integration & System Validation)

Duration: Aug 03 – Aug 12, 2026

- **Focus:** Connect user dashboards to live APIs, enforce access control, and eliminate functional bugs.
- **Frontend Integration:**
    - Connect the React frontend views with operational backend endpoints via standard routing wrappers.
    - Isolate user layouts dynamically, ensuring high-security separation between member dashboards and administrative panels.
- **System Testing:**
    - Execute full End-to-End flow routines tracking user operations from registration form inputs up to core AI feature workflows.
- **Bug Fixing:**
    - Identify and correct routing exceptions, data handling latency spikes, or cross-origin request anomalies.

- **User Testing & Feedback:**
    - Conduct iterative evaluation routines using mock data loops to ensure UI elements match the defined visual flows perfectly.
- **Documentation Updates:**
    - Track and detail system anomalies, performance adjustments, and error resolution files directly inside the repository logs.
- **Sprint Review & Deliverables:**
    - Fully linked, responsive software application displaying operational cross-page navigation and dynamic data binding.

Milestone 5 Plan (Final Submission Delivery)

Duration: Aug 13 – Aug 23, 2026

- **Focus:** Polish edge-case functionalities, record demonstration materials, and compile final engineering portfolio.
- **Final Testing:** Execute exhaustive pre-flight verification tests across all integrated modules to guarantee operational safety.
- **Final Documentation:** Polish the unified engineering report file, including step-by-step installation recipes for Windows, Mac, and Ubuntu environments.
- **Presentation & Demo:** Prepare high-impact slide presentations mapping the core lifecycle stages, engineering trade-offs, and GenAI features.
- **Video Recording:** Produce a high-resolution 5-10 minute presentation video walking through real user story validations and application workflow highlights.

Sprint Schedule

Sprint 1: Milestone 1 - User Requirement Discovery

- Categorize target user groups into primary, secondary, and tertiary tiers
- Perform stakeholder inquiries to uncover operational challenges without prescribing

features
- Archive verification artifacts (interview recording snippets)
- Draft structured SMART user stories (As a [role], I want [capability], so that [benefit])
- Assemble Milestone 1 submission PDF (user profiles, field analysis, and story backlog)
- Date: 23/06/2026 - 01/07/2026

Sprint 2: Milestone 2 - System Architecture & UI Prototyping

- Finalize visual branding, technology stack, and engineering toolchain
- Construct project schedule dependencies and master Gantt chart
- Configure GitHub Projects Kanban board and issue tracking workflows
- Model component interaction diagrams and database ER schemas
- Build connected frontend user screens with routing and RBAC guardrails
- Compile Milestone 2 PDF deliverable (system models, UI screenshots, schedule, and minutes)
- Date: 02/07/2026 - 22/07/2026

Sprint 3: Milestone 3 - Core API Development

- Identify and rank critical backend REST endpoints (6–8 core routes)
- Define OpenAPI/YAML contracts for request payloads and response structures
- Build data validation, JWT authentication middleware, and error handling
- Map backend endpoints directly to user story requirements
- Execute automated unit tests using pytest and track defects on GitHub
- Date: 23/07/2026 - 03/08/2026

Sprint 4: Milestone 4 - Full-Stack Integration & QA Testing

- Connect frontend React components to Flask backend REST APIs
- Implement and verify core end-to-end user workflows
- Design and execute test matrices (Input Data → Expected Outcome → Observed Execution)
- Triage and resolve full-stack integration bugs
- Update system technical documentation and run instructions in README.md
- Date: 04/08/2026 - 12/08/2026

Sprint 5: Milestone 5 - System Verification & Final Documentation

- Demonstrate full working application capabilities across all user roles
- Run final automated test suites (pytest) and capture coverage outcomes
- Document bug fixes and summarize stakeholder feedback

- Assemble consolidated Master PDF Report (all milestone outputs + UI screenshots)
- Prepare presentation slide deck and video demonstration outline
- Date: 13/08/2026 - 18/08/2026

Sprint 6: Final Submission & Release Packaging

- Perform final sanity checks across repository codebase and documentation files
- Finalize, render, and host official video demonstration clips
- Proofread master report and test setup execution scripts
- Package clean source code ZIP archive (excluding node_modules) with README.md
- Submit all final deliverables (Report PDF, Code ZIP, Video, Presentation Deck)
- Date: 19/08/2026 Onward

SCRUM Meetings

Scrum Meetings: Conducted on scheduled dates via Google Meet (9PM - 10PM)

Minutes from June 23, 2026 SCRUM Meeting:

- **Meeting Focus & Overview:** Conducted initial team setup, reviewed interview questions for drama society management, and interviewed Siddharth Sahu (Secretary of the IIT Drama Club). Analyzed operational challenges including script ideation, lack of

costume/venue budget, manual scheduling, and reliance on WhatsApp/Google Drive.
- **Key Decisions:** Decided to conduct and record primary user research, edit the recording (trimming pauses and applying a 1.25x speed multiplier), and focus software goals on solving real operational pain points for community drama clubs.
- **Assigned Tasks / Action Items:**
    - **Shibin Mohammed:** Handle video recording of the interview session, apply a 1.25x speed edit, and complete processing prior to the TA review.
    - **Jatin Bhardwaj:** Record backup audio via Mac native recorder, outline project strategy options (community/civic service), and organize meeting notes.
    - **Devi Vrinda A:** Update and share the interview question document in Discord, incorporating local event sharing and inventory management inquiries.
    - **Amit Kumar Patra:** Review ticket sales and inventory questions with the team to finalize the research framework.
    - **Bhabani Kundu:** Review interview recording logs and prepare to summarize user operational pain points.

Minutes from June 24, 2026 SCRUM Meeting:

- **Meeting Focus & Overview:** Defined system stakeholder classifications and established milestone submission protocols. Categorized primary users as club members/performers, secondary users as internal core committee maintainers and regulatory bodies, and tertiary users as sponsors/external venues.
- **Key Decisions:** Restricted research scope to the current interview data (no additional external interviews needed). Established that all team members must independently submit an Honor Code declaration, draft contributions in Markdown, and submit work via GitHub Pull Requests.
- **Assigned Tasks / Action Items:**
    - **Shibin Mohammed:** Conduct research and draft SMART user stories for primary club members and performers.
    - **Amit Kumar Patra:** Conduct research and draft SMART user stories for secondary administrative users and regulatory bodies.
    - **Devi Vrinda A:** Transform the raw interview audio into a structured Q&A transcript and document core user pain points.
    - **Jatin Bhardwaj:** Prepare the team Honor Code template, draft meeting minutes, and post task summaries to Discord.
    - **Bhabani Kundu:** Review user story requirements and assist in mapping pain points to system features.

Minutes from June 29, 2026 SCRUM Meeting:

- **Meeting Focus & Overview:** Transitioned from Milestone 1 to Milestone 2 planning, focusing on application architecture, dashboard separation, and Generative AI feature strategies.
- **Key Decisions:** Adopted a multi-page website layout featuring two distinct dashboard views: one for maintainers/secretaries and one for general members. Agreed to implement 6 to 7 AI-driven features (e.g., automated event summaries, role matching, audition quality checks). Designated the "Plays" page as a distinct entity from general

"Events" to showcase full scripts, cast lists, and technical details.

- **Assigned Tasks / Action Items:**
    - **Shibin Mohammed:** Draft design prompts and layout requirements for the Audition and Role Application pages.
    - **Amit Kumar Patra:** Collaborate with Jatin on selecting visual color palettes and explore UI generation tools (Stitch/Tailwind).
    - **Devi Vrinda A:** Draft design prompts for the About Page and research technical media evaluation tools on Hugging Face.
    - **Bhabani Kundu:** Draft design prompts for the Dashboard, Events page, and Gallery catalog.
    - **Jatin Bhardwaj:** Coordinate overall UI architecture, research audio/video quality model feasibility, and investigate sponsorship API options.

Minutes from July 01, 2026 SCRUM Meeting:

- **Meeting Focus & Overview:** Reviewed homepage wireframe outputs, evaluated design assets, and finalized the core frontend technology stack.
- **Key Decisions:** Reached a team consensus to adopt Shadcn UI and Tailwind CSS as the foundational design system to ensure visual uniformity across reusable components. Reaffirmed building frontend components with React and TypeScript (.tsx).
- **Assigned Tasks / Action Items:**
    - **Shibin Mohammed:** Initialize the React/Shadcn development environment and begin coding the frontend event and audition pages.
    - **Amit Kumar Patra:** Document the design system setup (Shadcn/Tailwind configuration) and draft secondary page components.
    - **Jatin Bhardwaj:** Construct the master Gantt chart to track Milestone 2 project timelines.
    - **Devi Vrinda A:** Request updated, high-resolution media assets and photos from the client's Google Drive.
    - **Bhabani Kundu:** Review frontend design components to align upcoming database field specifications.

Minutes from July 08, 2026 SCRUM Meeting:

- **Meeting Focus & Overview:** Audited frontend component organization, defined database schema fields, and established strict Role-Based Access Control (RBAC) boundaries.
- **Key Decisions:** Organized codebase directory structure by isolating the sidebar component and creating dedicated subfolders under src/pages/ for member and maintainer views. Decided to store script data as Markdown text within SQLite. Removed the separate announcements database in favor of an integrated notification system.
- **Assigned Tasks / Action Items:**
    - **Shibin Mohammed:** Develop member dashboard components (MemberDashboard.tsx), incorporating live rehearsal countdowns, RSVP state toggles, and contextual schedule notes.
    - **Amit Kumar Patra:** Develop maintainer dashboard components in a separate subfolder and implement database tables for plays, scripts, and locations.

    - **Jatin Bhardwaj:** Build the public landing page with login access toggles and perform code reviews on open PRs.
    - **Devi Vrinda A:** Define database schema fields for volunteers vs. participants and post dummy database code to GitHub issue comments.
    - **Bhabani Kundu:** Refine SQLite database models (member profiles, events, finance) and format field definitions in Markdown.

Minutes from July 11, 2026 SCRUM Meeting:

- **Meeting Focus & Overview:** Established backend technology selection, authentication mechanisms, and API routing logic.
- **Key Decisions:** Formally selected Flask as the backend framework. Adopted JSON Web Tokens (JWT) and secure password hashing for authentication instead of third-party OAuth/Google Login. Implemented automated backend routing to direct users to either member or maintainer dashboards based on authenticated user roles.
- **Assigned Tasks / Action Items:**
    - **Jatin Bhardwaj:** Initialize and upload the baseline Flask backend repository structure and database connection routes.
    - **Bhabani Kundu:** Implement JWT authentication logic, secure session handling, and backend login endpoints.
    - **Devi Vrinda A:** Collect real club member data/photos and build database populator scripts.
    - **Shibin Mohammed:** Align member frontend API calls with Flask backend authentication endpoints.
    - **Amit Kumar Patra:** Align maintainer frontend API calls and research payment gateway integration for patron circle funding.

Minutes from July 21, 2026 SCRUM Meeting:

- **Meeting Focus & Overview:** Finalized research responsibilities for the 5 core Generative AI features and defined technical scope boundaries.
- **Key Decisions:** Restricted the "Audition Sanity Check" feature strictly to technical audio/video verification (resolution, audio clarity, missing frames) rather than subjective acting performance evaluation. Agreed to prioritize research, documentation, and interface screenshots over full live model training for the upcoming deadline.
- **Assigned Tasks / Action Items:**
    - **Shibin Mohammed:** Lead research on the "Rehearsal Bot / Q Partner & Line Reader" feature (audio streaming, speech-to-text with word-level timestamps).
    - **Amit Kumar Patra:** Lead research on the "AI Character Generator & Manuscript Assistant" feature.
    - **Devi Vrinda A:** Lead research on the "Audition Sanity Check" tool using Hugging Face audio/video verification models.
    - **Jatin Bhardwaj:** Lead research on the "Sponsor Key Point Filter" tool for extracting portfolio highlights.
    - **Bhabani Kundu:** Lead research on the "Event Detail Summarizer" AI feature.

Minutes from July 21, 2026 SCRUM Meeting:

- **Meeting Focus & Overview:** Finalized repository file organization, reviewed AI prototype progress, and assigned final Milestone 2 report packaging tasks.
- **Key Decisions:** Established the meeting minutes format as concise, pointwise summaries including dates and task assignments. Standardized the final project submission package to contain separate folders for: (1) Consolidated Deliverable PDF, (2) Clean Frontend Source Code ZIP (excluding node_modules), and (3) Combined Milestone 1 & 2 documentation archives.
- **Assigned Tasks / Action Items:**
    - **Shibin Mohammed:** Compile the official Scrum Meetings Schedule and Minutes Markdown file and submit Pull Request.
    - **Bhabani Kundu:** Complete the Project Schedule documentation section (referencing pages 8–13) and export finalized Gantt charts.
    - **Devi Vrinda A:** Generate final system architecture and database entity-relationship (ER) class diagrams.
    - **Amit Kumar Patra:** Package the clean frontend source code into a ZIP file (removing node_modules) and integrate AI frontend preview components.
    - **Jatin Bhardwaj:** Conduct final PR code reviews, verify clean TypeScript compilation, and assemble the consolidated Milestone 2 PDF Report.

PROJECT SCHEDULING TOOLS

To manage the development of the **Aayam** platform effectively, visual task management and scheduling tools were utilized to organize work across different phases, align team member responsibilities, and ensure timely completion of project milestones.

Summary of Tools Used

Tool	Category	Primary Function	Key Value Provided
Kanban Board (GitHub Projects)	Visual Task Management	Workflows, ticket statuses, and task allocation	Provides real-time visibility into task progress, active bottlenecks, and individual bandwidth.
GitHub Issues & PRs	Bug & Code Tracking	Issue logging, bug tracking, and code reviews	Ensures full traceability between user stories, code commits, pull requests, and bug resolutions.
Gantt Chart (Excel / Matplotlib)	Schedule & Timeline Visualization	High-level schedule and dependency mapping	Displays start/end dates, milestone deadlines, and parallel development workstreams.
Stitch	UI/UX Prototyping & Design Systems	High-fidelity wireframing and UI variant generation	Early visual validation of dashboard designs against Milestone 1 user requirements.
Draw.io	System Architecture Modeling	Class diagrams and database ER diagrams	Visual modeling of domain classes and database schema relationships (aayam.db).
Discord & Google Meet	Team Communication & SCRUM	Async daily standups and weekly meetings	Enables fast resolution of blockers and documentation of meeting minutes.

Tool Descriptions & Workflow Integration

- **Kanban Board (Task Management):** We primarily used a Kanban board to manage and track task progress throughout the project lifecycle. Tasks were categorized into columns representing development stages (Backlog, To Do, In Progress, In Review, Done). This structure enhanced transparency across frontend (React/Tailwind) and backend (Flask/SQLite) workstreams, ensuring all team members had clear visibility into task ownership.
- **GitHub Issues (Bug & Issue Tracking):** For issue tracking during development and testing, GitHub Issues was used to log bugs, blockers, and feature enhancements. Each issue was assigned priority tags (P0, P1, P2), linked to relevant user stories (e.g., practice room booking, script builder AI integration), and tracked until resolution to maintain a complete audit log.

- **Gantt Chart (Project Timeline):** A Gantt chart was created to represent the complete project schedule across all milestones and sprints from June to August 2026. It provided a high-level view of deadlines, critical path dependencies, and overlapping tasks, ensuring that planning, UI design, API development, integration, and testing were completed on schedule.
- **Stitch (UI Prototyping & Design Systems):** We utilized Stitch for UI/UX prototyping, design system generation, and creating screen variants across Aayam's core user roles (Member Dashboard, AI Rehearsal Studio, Production Office, Maintainer Control Panel). This enabled rapid visual validation of UI layouts against Milestone 1 user stories prior to React implementation.
- **Discord & Google Meet (Team Communication & SCRUM):** We used Discord text/voice channels for daily asynchronous standup updates, real-time developer coordination, and quick technical discussions, alongside scheduled Google Meet sessions for weekly SCRUM progress reviews and milestone demonstrations.

Expected Outcomes & Benefits

The integration of these scheduling tools enabled:

- **Efficient Tracking:** Seamless monitoring of task progress across individual sprints.
- **Workflow Clarity:** Clear visual representation of active workstreams and task handoffs.
- **Proper Issue Documentation:** Systematic logging and resolution of software bugs and technical debt.
- **On-Time Delivery:** Disciplined adherence to project milestones and final submission deadlines.

GANTT CHART & SCHEDULE TIMELINE

Overview & Description

The Gantt chart represents the complete project timeline from **Milestone 1 (User Requirement Analysis)** to **Milestone 5 (Final Submission)**. It provides a visual representation of all major phases, including planning, system design, API development, frontend-backend integration, testing, and final delivery.

The project schedule is structured into sequential milestones and overlapping sprints:

- **Milestone 1 (User Requirement Analysis):** Focuses on identifying user requirements

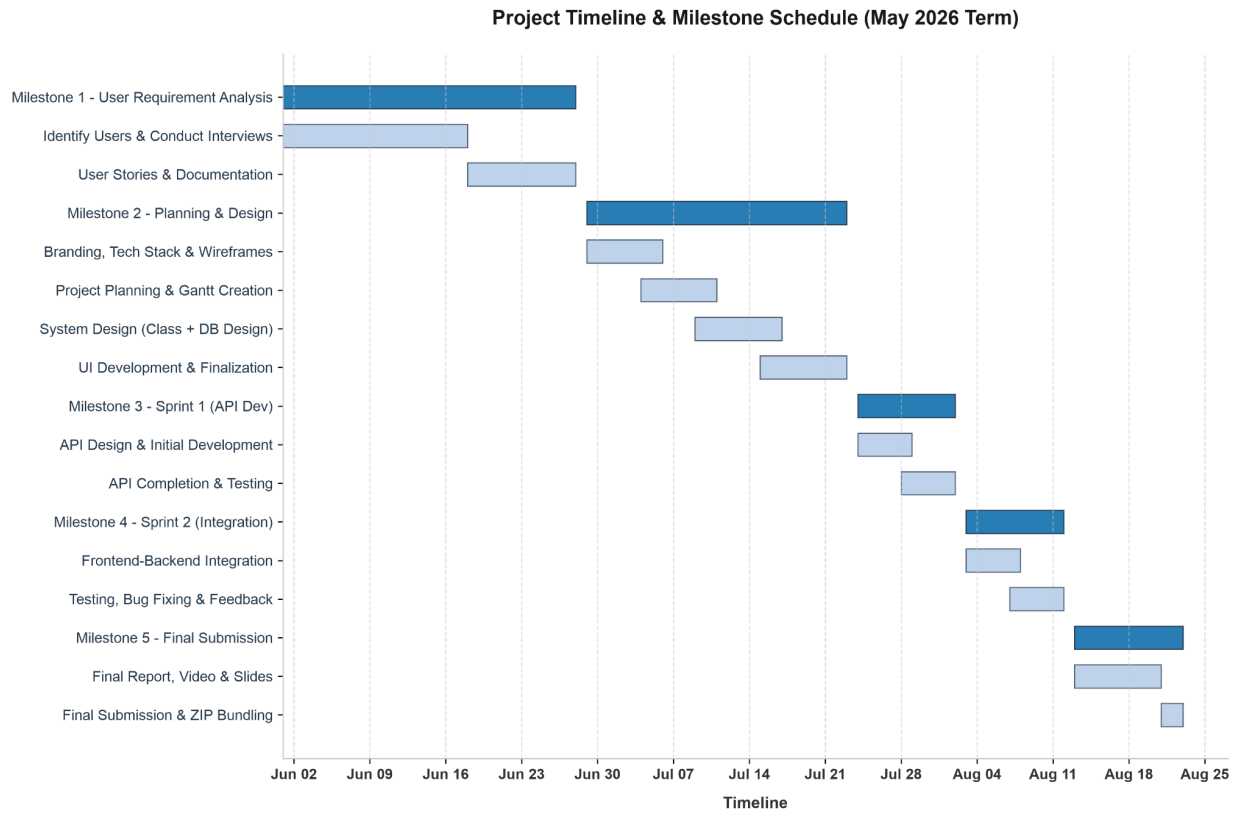
through interviews with Club Secretary Siddharth Sahu, active performers, prospective members, and sponsors, leading to 13 SMART user stories.
- **Milestone 2 (Planning & Design):** Covers planning and design activities, including project scheduling, Gantt chart creation, system architecture (class diagrams, ER schema for *aayam.db*), and frontend UI prototyping.
- **Sprint 1 / Milestone 3 (API Development):** Focuses on core backend API design (Flask REST endpoints), database models, and automated testing with Pytest.
- **Sprint 2 / Milestone 4 (Testing & Integration):** Focuses on connecting React frontend components to backend APIs, conducting end-to-end user flow testing, and fixing integration bugs.
- **Milestone 5 (Final Submission):** Involves final testing, report compilation, video demonstration recording, and final ZIP packaging.

2.2 Detailed Gantt Chart Task Table

Below is the complete schedule task table detailing task names, start dates, and end dates as derived from the updated Gantt chart (`gantt_chart_updated.png`).

	Task Name	Start Date	End Date
1.0	**Milestone 1 - User Requirement Analysis**	**02-06-2026**	**28-06-2026**
1.1	Identify Users & Conduct Interviews	02-06-2026	18-06-2026
1.2	User Stories & Documentation	18-06-2026	28-06-2026
2.0	**Milestone 2 - Planning & Design**	**28-06-2026**	**23-07-2026**
2.1	Branding, Tech Stack & Wireframes	28-06-2026	05-07-2026
2.2	Project Planning & Gantt Creation	03-07-2026	10-07-2026
2.3	System Design (Class Diagram + DB Design)	08-07-2026	15-07-2026
2.4	UI Development & Linking	14-07-2026	23-07-2026
2.5	Documentation & Finalization (Milestone 2)	18-07-2026	23-07-2026
3.0	**Sprint 1 - API Development (Milestone 3)**	**23-07-2026**	**02-08-2026**
3.1	API Design & Initial Development	23-07-2026	29-07-2026

	Task Name	Start Date	End Date
3.2	API Completion & Testing	28-07-2026	02-08-2026
4.0	**Sprint 2 - Integration & Testing (Milestone 4)**	**02-08-2026**	**12-08-2026**
4.1	Frontend-Backend Integration	02-08-2026	08-08-2026
4.2	Testing, Bug Fixing & Feedback	07-08-2026	12-08-2026
5.0	**Milestone 5 - Final Submission**	**12-08-2026**	**20-08-2026**
5.1	Final Report, Video & Slides Preparation	12-08-2026	18-08-2026
5.2	Final Submission & ZIP Bundling	18-08-2026	20-08-2026



Key Features & Critical Path Analysis

- **Timeline Clarity:** Clearly outlines start dates, end dates, and target milestones for the

Aayam project.
- **Parallel Workstream Execution:** Represents concurrent tasks (e.g., UI development running alongside database schema design and SQLite seed data preparation).
- **Phase Dependency Mapping:** Enforces clear task sequence: Design → API Development → Integration → Testing → Final Delivery.
- **Time Management:** Prevents bottlenecks by allocating dedicated buffer periods for integration and testing.

PROJECT TRACKING MECHANISM (KANBAN BOARD)

Overview

To effectively manage and track project progress for **Aayam**, a **Kanban board** was used as the primary project tracking tool. It provided visual representation of tasks across different stages of development, ensuring transparency and efficient workflow management across team members (Jatin, Amit, Shibin, Devi Vrinda, Bhabani).

Board Column Structure

The Kanban board was structured into 5 columns representing task status:

- **Backlog:** Tasks planned for future phases (e.g., confidential grievance portal, external workshop booking engine) but not yet started.
- **To Do:** Priority tasks refined and ready to be picked up by team members.
- **In Progress:** Tasks currently actively being developed (WIP limits applied to maintain focus).
- **In Review:** Tasks undergoing code review, UI verification, or document proofreading.
- **Done:** Completed tasks that meet the Definition of Done (DoD).

Ticket Governance & Metadata

Each task was represented as a card containing:

- **Task Title & Description:** Clear summary of task scope and acceptance criteria.
- **Priority Level:** Classified as P0 (Critical/Blocker), P1 (High Priority), P2 (Normal), or P3 (Low).

- **Assignee:** Single point of accountability assigned to each card (e.g., Jatin for Frontend, Bhabani for Backend API, Shibin for Code Review).
- **Milestone Tagging:** Explicit labels such as [M2][Frontend], [M3][API], or [M4][Integration].
- **GitHub Integration:** Direct linking between Kanban cards, GitHub Issues, and Pull Requests.

Outcomes Achieved

- **Efficient Task Tracking:** Seamless visibility into active development items.
- **Enhanced Coordination:** Improved team communication and individual accountability.
- **WIP Control:** Prevention of task overload by limiting work-in-progress items.
- **Timely Deliverables:** Reliable progress tracking resulting in on-time milestone submissions.

FRONTEND INTERFACE SCREENS

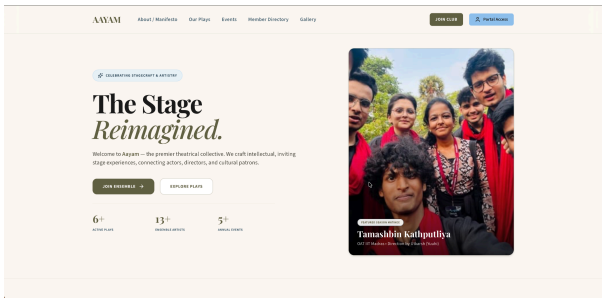
The frontend of **Aayam (IIT BS Drama Society Platform)** is designed to support multiple user roles (Club Members, Core Leadership, New Applicants, System Maintainers, Sponsors, and External Partners), each with a dedicated dashboard tailored to their specific operational responsibilities.

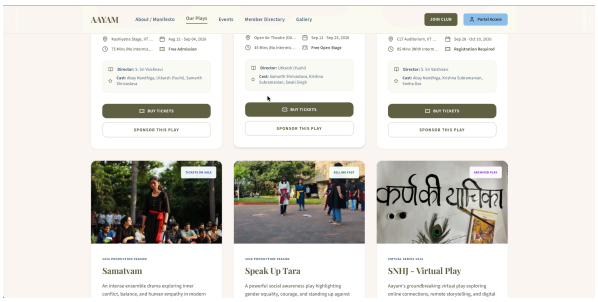
Below are the comprehensive text specifications for the core frontend modules and interface screens developed in React and Tailwind CSS.

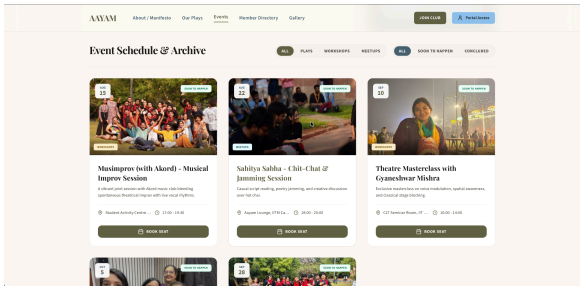
Landing Page & Public Cultural Portal

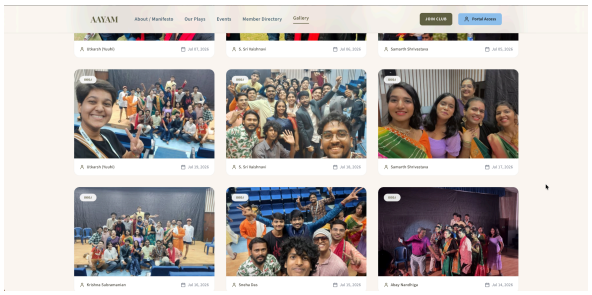
Designed as the public digital storefront for the Aayam Drama Society to engage prospective members, showcase past productions, and publish upcoming event schedules.

- **Hero Banner & Production Spotlight:** Dynamic banner highlighting flagship productions (e.g., Matinee Club, annual stage plays) with trailer video embeds and ticket booking CTAs.
- **Public Event & Audition Announcement Feed:** Visual cards showcasing upcoming street plays, stage auditions, and workshop schedules with date, venue, and eligibility criteria.
- **Society Impact Statistics:** Performance indicators showcasing total active members, past shows conducted, total audience reach, and awards won.
- **Quick Access Navigation:** One-click navigation routing users to Member Login, New Applicant Registration (JoinClub.tsx), Production Office, and Sponsor Portal.





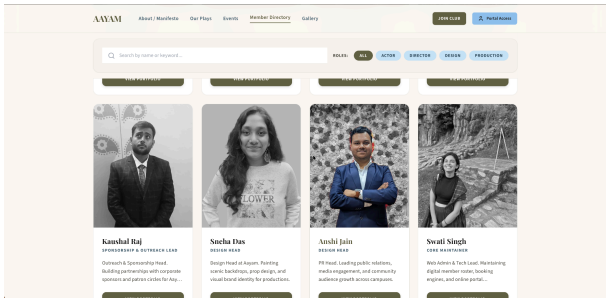


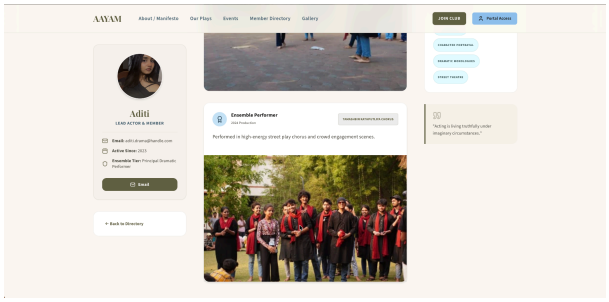


Join Club & Onboarding Portal

A guided digital recruitment wizard designed to streamline member onboarding and eliminate physical paper application forms.

- **Interest Track Selection:** Multi-track selection interface allowing applicants to choose preferred sub-groups (Acting, Scriptwriting, Directing, Stage Management, Costume & Makeup, Sound & Lighting).
- **Portfolio & Media Sample Upload:** Drag-and-drop file uploader supporting headshots, voice clips, monologue videos, and portfolio PDFs.
- **Audition Slot Picker:** Interactive calendar allowing applicants to reserve specific audition time slots without scheduling conflicts.
- **Real-Time Application Status Tracker:** Visual progress bar allowing applicants to monitor their application status (Submitted, Under Review, Audition Scheduled, Selected, Onboarded).

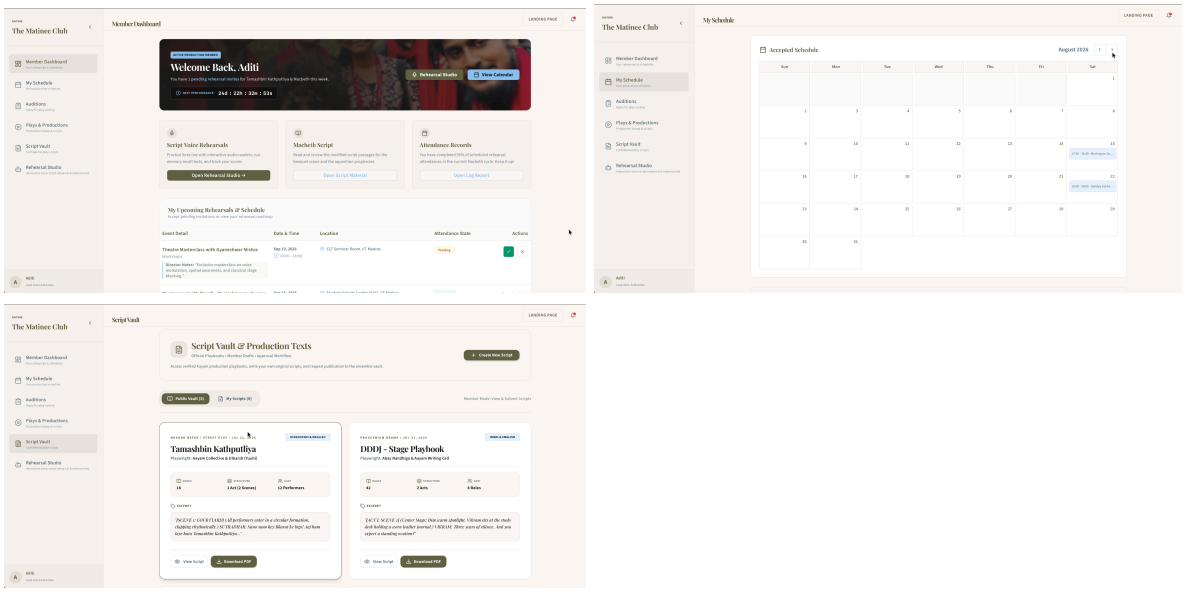




Member Dashboard & Personal Workspace

A personalized operational hub for active drama club members, cast, and crew.

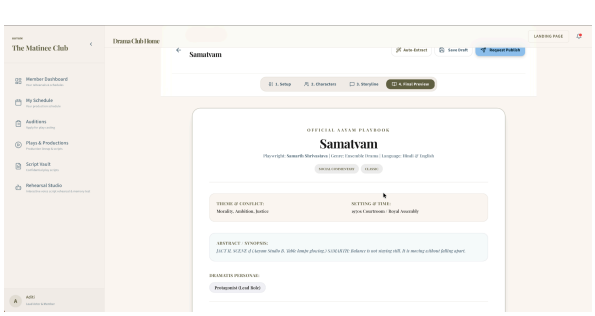
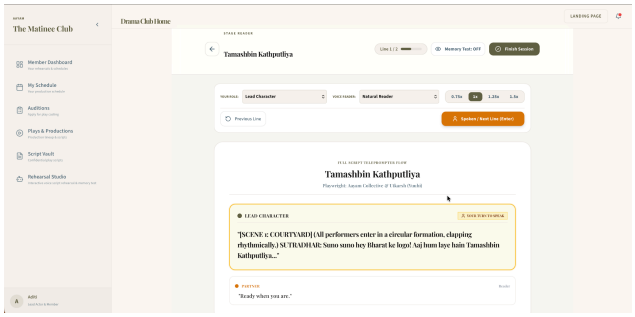
- **Personalized Schedule & Call Times (MySchedule.tsx):** Timeline feed displaying assigned rehearsal timings, call times, practice room reservations, and assigned production duties.
- **Rehearsal Room & Space Reservation Engine:** Calendar booking interface enabling members to reserve open practice rooms up to 7 days in advance with automated double-booking prevention.
- **Script & Play Access Library (MemberPlays.tsx, MemberScripts.tsx):** Digital library providing access to approved play scripts, character breakdowns, and rehearsal audio notes.
- **Artistic Profile & Skill Portfolio (ViewProfile.tsx, EditProfile.tsx):** Profile management screen allowing members to maintain verified records of past roles, artistic skills, availability, and emergency contacts.



AI Rehearsal Studio & Script Builder

An innovative GenAI-powered creative workspace designed for scriptwriting and line practice.

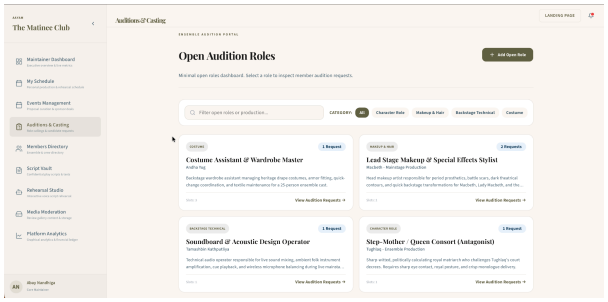
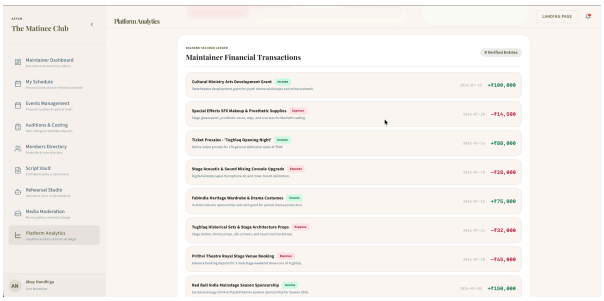
- **Interactive Script Builder (ScriptBuilder.tsx):** Professional script formatting editor supporting character dialog tagging, scene descriptions, and act structuring.
- **AI Dialog & Scene Generation:** Gemini AI-assisted writer helping playwrights generate character dialogues, scene transitions, and creative plot twists.
- **AI Line Practice & Partner Simulation (AiRehearsalStudio.tsx):** Voice/text AI practice partner that reads opposing character lines in real time so actors can practice line delivery independently.

Production Office & Backstage Pass

An administrative control center designed for Club Leadership (Secretary Siddharth Sahu, Directors, Event Organizers).

- **Casting & Audition Management Console:** Interface allowing leaders to review audition recordings, rate applicant performances, and assign play roles with one click.
- **Stage Equipment Issue & Inventory Tracker:** Digital checkout log for stage gear (microphones, sound amplifiers, costumes, props) tracking item issuance, borrower member IDs, and return status.
- **Production Call Sheet Generator:** Workspace tool to generate and distribute daily production call sheets to cast and crew members.
- **Budget & Expense Tracking:** Expense logging table enabling organizers to record production expenses and upload digital receipt images for reimbursement processing.

Maintainer & Admin Control Panel

A comprehensive system administration suite for technical leads, system maintainers, and

regulatory oversight representatives.

- **All-Members Directory & RBAC Governance (AllMembersDirectory.tsx):** Comprehensive user database allowing administrators to manage user accounts, update Role-Based Access Controls (RBAC), and trigger annual committee account handovers.
- **Compliance & Financial Audit Ledger (ComplianceLedger.tsx):** View-only ledger for institute authorities (Student Affairs Office, Faculty Coordinators) to audit financial transactions, event safety plans, and facility usage logs.
- **Confidential Misconduct & Grievance Log:** Sequestered portal allowing students to submit confidential safety or behavioral grievances directly to institute authorities.
- **Platform Analytics & Media Moderation (PlatformAnalytics.tsx, MediaModeration.tsx):** Analytics dashboard tracking system usage, active user retention, and media content moderation queues.

